QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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Contents

Document Control 3

PRELIMINARY ROUND (Test Strategy & Planning) 3

1. Scope & Requirements Traceability 3

2. Risk Assessment & Mitigation Strategy 4

3. Test Environment & Execution Strategy 5

4. CI/CD Release Thresholds & Automation Gates 6

5. Test Case Specifications (Drafts) 6

6. AI Output & Boundary Testing (Drafts) 7

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Document Control

Field	Detail
System Under Test (SUT)	UMHackathon 2026 - Team Midas UTeM
Team Repo URL	https://github.com/hecctarr/InsureFlow.git
Project Board URL	[Insert link to your Jira/Trello Projects board]
Live Deployment URL (optional for preliminary round)	-

Objective:

The primary objective is to ensure that **InsureFlow Pro** can reliably analyze unstructured medical data and complex insurance policy frameworks to generate accurate, actionable claim verdicts using AI-driven reasoning under varying system conditions (including API timeouts).

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

1.1 In-Scope Core Features

- Multi-Modal Claim Intake: AI processing of digital PDFs and handwritten medical photos.
- AI Decision Orchestrator: GLM-4 reasoning engine for cross-referencing documents.
- Autonomous Policy Matching: Matching medical jargon to legal policy clauses.
- Dynamic Payout Calculation: Automated math for deductibles and co-insurance.
- Actionable Communication: Generation of professional claim email drafts.

1.2 Out-of-Scope

- User authentication & full account management.
- Integration with live banking payment gateways.
- Mobile application native version.
- App Theme customization.

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2. Risk Assessment & Mitigation Strategy

Technical Risk	Likelihood	Severity	Risk Score(L×S)	Mitigation Strategy	Testing Approach
AI Incorrect Decision (Hallucination)	2	5	10	Use structured prompting (JSON output enforcement) and low Temperature (0.1).	Run 6 distinct medical scenarios (Ahmad, Siti, Chong, Tan, etc).
API Gateway Failure (504 Timeout)	4	4	16	Implementation of a **Local Heuristic Agent** as a fail-safe mechanism.	Simulate network failure/offline mode during the demo.
OCR Extraction Failure	3	3	9	Leverage Multi-modal GLM-4 vision capabilities for handwriting.	Test with low-quality photos of medical discharge notes.

Risk Assessment Scoring Criteria:

Likelihood (1-5)		Severity (1-5)	
1	Rare	1	Impact is Negligible
2	Unlikely	2	Impact is Minor
3	Possible	3	Moderate is Minor
4	Likely	4	Major Impact
5	Almost Certain	5	Critical Failure of the system

Risk Score = Likelihood × Severity

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Risk Level Reference:

Risk Score	Risk Level	Recommended Action
1 – 5	Low	Monitor only. Acceptable Risks
6 - 10	Medium	Mitigate + Testing
11 – 15	High	Must need mitigating and through testing is required.
16 – 25	Critical	Priority is Highest. Need extensive level of testing

3. Test Environment & Execution Strategy

- **Unit Testing**
  - **Scope:** PDF extraction function, Regex pattern-matching.
  - **Execution:** Manual testing via Python scripts during development.
  - **Pass Condition:** Extraction returns clean, capitalized strings for reasoning.
- **Integration Test**
  - **Scope:** End-to-end data flow (Upload → Extract → Z.AI → Dashboard).
  - **Workflow:** Real API calls to Ilmu.ai integrated with Streamlit UI.
  - **Pass Condition:** Metrics (Payout, Deductible) match the document data perfectly.
- **Test Environment (CI/CD Practice)**
  - **Local Testing:** Manual testing on localhost:8501.
  - **CI:** GitHub repository version control and integrity checks.
- **Regression Testing & Pass/Fail Rules**
  - **Execution:** Re-testing all 6 core scenarios after every logic change.
  - **Pass Condition:** Payout values must be 100% consistent with policy rules.
- **Test Data Strategy**

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- **Manual:** 6 curated scenarios (Ahmad, Siti, Chong, Bakri, Vikram, Tan).
- **Edge:** Testing with "Pre-existing condition" reports to verify REJECTION logic.

4. CI/CD Release Thresholds & Automation Gates

4.1 Integration Thresholds (Merging to Main)

- Zero syntax/build errors.
- 100% manual test pass rate for the 6 core scenarios.
- No exposed API keys in the public repository.

4.2 Deployment Thresholds (Ready for Demo)

Checks	Requirements	Project Result	Status
Regression Test	100% Pass	100%	**Passed**
AI Output Accuracy	>= 90%	96%	**Passed**
Uptime (Fail-Safe)	100% (No Error 504)	100%	**Passed**
Security	Session State Purge	Clean	**Passed**

5. Test Case Specifications (Drafts)

Test ID	Type	Description	Steps	Expected Result	Actual Result
TC-01	Happy Case	Full claim flow (Surgical)	Upload PDF -> Initiate Flow	Eligible; RM 4,800 Payout.	**Passed**
TC-02	Negative	Rejected Case (Mdm Tan)	Upload Chronic PDF -> Initiate	**REJECTED** status; Pre-existing flag.	**Passed**
TC-03	NFR	Concurrent API Load	Trigger flow while API is slow	System switches to **Agentic**	**Passed**

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				Local Engine	
TC-04	NFR	Multi-modal OCR	Upload photo of handwriting	AI identifies diagnosis from image	**Passed**

6. AI Output & Boundary Testing (Drafts)

6.1 Prompt/Response Test Pairs

Test ID	Prompt Input	Expected Output (Criteria)	Status
AI-01	Analyze Uncle Chong (Day-care)	Payout: RM6,000; Deductible: RM0.00.	**Pass**
AI-02	Analyze Siti (ACL Injury)	Deductible: 10% Co-insurance.	**Pass**

6.2 Oversized/Larger Input Test

- **Maximum Input Size:** ~2,000 Tokens per document.
- **Input used while testing:** ~5,000 words (Large policy booklet).
- **Actual Behaviour:** System successfully truncates/chunks data for reasoning. (Status: Pass)

6.3 Adversarial/Edge Prompt Test

- **Input:** Uploading a picture of a car instead of a medical report.
- **Expected Result:** Agent responds with "Unknown Patient/Diagnosis" safely without hallucinating a claim.

6.4 Hallucination Handling

The system mitigates hallucinations by:

- **Grounded Logic:** AI is restricted to only use data found within the uploaded files.

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- **Structure Enforcement:** Enforcing a JSON response format to prevent AI from "storytelling" false data.
- **Transparent Reasoning:** Providing a "Logic Audit" tab so humans can verify the AI's thought process.

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